The Rank-Nine Determinant Theorem

Full mathematical foundations, exact finite reduction, and a detailed audit

Confidential research working document

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Status and scope. This document explains the computer-assisted result $e_9(6) = 21870$, audits the mathematical reduction and its exact certificates, and distinguishes certificate validation from exhaustive regeneration. The precise replay status is recorded in Section 15. The audit is internal; an independent coauthor or external verification has not been represented as completed. The supplied source manuscript is confidential. Its ambient embeddings, kissing numbers, and separate cluster constructions are outside the checks reported here.

Abstract

We study positive-definite even integral lattices of rank nine and minimum at least six, where "minimum" always means squared Euclidean length. The claimed exact optimum is determinant 21870. An explicit lattice attains this value. To prove the lower bound, a hypothetical counterexample is cut repeatedly by shortest primitive dual vectors. Hermite bounds, Mordell's inequality, determinant congruences, and a lifting inequality place this saturated flag in finite determinant windows. Complete exact one-vector extension enumeration reaches 2235 labelled rank-seven forms; 96 survive the lifting bound and are identified with six representatives. Exact discriminant-class minima close their possible rank-eight continuations.

The exposition supplies the linear algebra, lattice theory, arithmetic algorithms, completeness arguments, matrices, and numerical derivations needed to understand this proof. It also corrects the scaling notation for the determinant-8192 lattice, makes the parent-minimum hypothesis in the layer criterion explicit, and identifies an omitted zero-residue layer in one generic helper. That helper defect does not change the particular branch conclusions, for a proved numerical reason. The result determines an abstract rank-nine determinant minimum and its consequence for the bare dimension-39 cross-section construction. Uniqueness, full classification, rank ten, and cluster optimality remain separate questions.

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1 The original problem and the exact claim being audited

The source is the supplied confidential manuscript by I. Dorofeev, X. Sun, and C. Wang, *Optimal Extensions of Cross-Sections: Sphere Packings in Dimensions 38 to 43*. The relevant passage in Section 7, page 13 reads as follows. Only line breaks and mathematical typesetting have been changed.

> Table 4 localizes what is left for the bare cross-section route and separates unconditional ceilings from current-record benchmarks. In dimensions 40, 41 and 42 the bare route is closed: the sections of Corollary 11 are optimal among bare cross-section packings. This does not rule out antipode clusters over these or other sections. In dimension 43 the bare route is also closed, while the cluster route is essential for our improvement and may admit further gains.
>
> The interesting bare cases are dimensions 39 and 38. In dimension 39, any rank-9 section with determinant below 21870 would improve our present packing. The current published rank-9 lattice record corresponds to the benchmark 19683: under the published-literature convention used here, a determinant below that is not excluded by an optimality theorem, but would simultaneously produce a new 9-dimensional lattice record. Thus $[19683, 21870)$ is the portion of the search interval that could improve dimension 39 without necessarily improving that lower-dimensional record. If the recent dimension-9 preprint cited above is correct, then determinants below 19683 are excluded and this interval is the full rigorous search window. Likewise, for dimension 38 one seeks determinant below 54675. The current rank-10 lattice record gives the exact benchmark $3^{11}/4 = 44286.75$; because section determinants are integral, $\det K \le 44286$ would also improve the rank-10 lattice record, while $44287 \le \det K < 54675$ could improve dimension 38 without doing so.
>
> Any such section inside an even ambient lattice must itself be an even integral lattice of minimum at least 6, which is a real constraint: at the rank-9 benchmark $19683 = 3^9$ the natural candidate, the rescaled laminated lattice $\sqrt{\frac{3}{2}}\,\Lambda_9$, is not integral, so a rank-9 section of that determinant would have to be some other lattice. We do not know which even integral lattices of minimum 6 occur in ranks 9 and 10, and determining their smallest determinant would sharpen these searches. The current lattice records alone do not provide rigorous determinant floors in these ranks; the recent dimension-9 preprint, if correct, supplies the rank-9 floor. We leave the corresponding integral-section questions open.

The statements about the literature in this quotation are part of the historical source text. The proof developed below does not use the dimension-nine preprint mentioned there. Its rank-nine Hermite estimate is the unconditional bound derived from dimension eight by Mordell's inequality.

Definition 1.1. For a positive integer r, put

$$e_r(6) = \inf\{\det L : L \text{ is positive definite, even and integral, } \operatorname{rk} L = r,\ \min L \ge 6\}.$$

The set is nonempty, since $\sqrt{6}\mathbb{Z}^r$ belongs to it. Its determinants are positive integers, so the infimum is a minimum.

Theorem 1.2 (the audited rank-nine statement)**.** *Every positive-definite even integral lattice L of rank nine with $\min L \ge 6$ satisfies $\det L \ge 21870$. The explicit Gram matrix in Section 13 attains this bound. Consequently*

$$\boxed{e_9(6) = 21870.}$$

The lower-bound proof uses the complete finite computations specified below, together with the stated classical Hermite theorem in ranks at most eight.

This quantifies over *all abstract* even integral lattices of rank nine. It therefore gives a necessary lower bound for every rank-nine section inside an even ambient lattice of minimum

six. A converse statement that every such abstract lattice embeds in a particular ambient lattice would be a different assertion and is not used.

The theorem closes the rank-nine minimum-determinant part of the quoted question. It does not give a list of all rank-nine lattices, prove that the minimizer is unique, or settle the rank-ten question in the same passage.

2 Lattices, Gram matrices, and normalization

2.1 Why the determinant is the square of the covolume

Let V be an r-dimensional Euclidean space and let $b_1, \dots, b_r$ be a real basis. A lattice is the discrete subgroup

$$L = \mathbb{Z}b_1 + \cdots + \mathbb{Z}b_r.$$

In an orthonormal coordinate system, write B for the matrix whose columns are the b_i. The Gram matrix is

$$G = B^{\mathsf{T}}B, \qquad G_{ij} = \langle b_i, b_j \rangle.$$

For an integer coordinate column x, the associated lattice vector is Bx, and

$$\|Bx\|^2 = x^{\mathsf{T}}Gx.$$

The volume of a fundamental parallelepiped is $\operatorname{covol} L = |\det B|$. Thus

$$\det L := \det G = (\operatorname{covol} L)^2.$$

This convention matters: a bound on the Gram determinant is the square of the corresponding covolume bound.

Another lattice basis has the form BX, where $X \in \mathrm{GL}_r(\mathbb{Z})$, equivalently X is integral and $\det X = \pm 1$. The new Gram matrix is $X^{\mathsf{T}}GX$. Its determinant is unchanged. Two Gram matrices A, R define integrally isometric lattices precisely when an integer matrix X satisfies

$$X^{\mathsf{T}}RX = A, \qquad |\det X| = 1.$$

The equality alone, without unimodularity or a same-determinant argument, can describe a proper sublattice and is not generally an isometry certificate.

2.2 Minimum, integrality, evenness, and scaling

The minimum is

$$\min L = \min_{0 \neq x \in \mathbb{Z}^r} x^{\mathsf{T}}Gx.$$

Throughout this note it is a *squared* length. Positive definiteness implies that a bounded ball meets the lattice in finitely many points, so this positive minimum is attained.

The lattice is integral if G has integer entries. It is even if, in addition, its diagonal entries are even. Indeed,

$$x^{\mathsf{T}}Gx = \sum_i G_{ii}x_i^2 + 2\sum_{i<j} G_{ij}x_ix_j.$$

Thus even diagonal entries suffice, and taking $x = e_i$ proves their necessity. A positive-definite even integral lattice has positive norms in $2\mathbb{Z}$; therefore proving $\min L \geq 6$ amounts to excluding norms 2 and 4. An exact enumeration through norm 5 is a convenient single check.

If all vectors are multiplied by $c > 0$, then

$$G(cL) = c^2G(L), \quad \min(cL) = c^2 \min L, \quad \det(cL) = c^{2r} \det L.$$

For example, $\sqrt{3}E_7$ has Gram matrix three times a root Gram matrix of E_7. If C is such a root Gram matrix, $4C^{-1}$ is the Gram matrix of $2E_7^*$, not of $4E_7^*$. The latter would have Gram matrix $16C^{-1}$. This resolves a notation error in the earlier draft and in a legacy transcript; the actual integer matrix used in the computation is unchanged.

2.3 Indices and saturated sublattices

If $J \subset L$ has the same rank and index h, a basis matrix of J is BX with $|\det X| = h$. Hence

$$\det J = h^2 \det L. \tag{1}$$

For a lower-rank sublattice $J \subset L$, its saturation is

$$J^{\mathrm{sat}} = L \cap \operatorname{span}_{\mathbb{Q}} J.$$

The following conditions are equivalent: $J = J^{\mathrm{sat}}$; L/J is torsion-free; and a basis of J extends to a basis of L. The equivalence follows from Smith normal form: the torsion invariant factors in a presentation of the inclusion must all equal one. A sublattice with these properties is called saturated or primitive.

Saturation is essential in the proof. If J has corank one and is saturated, then $L/J \cong \mathbb{Z}$, so one vector extends a basis of J to a basis of L. Without saturation, a one-vector extension does not describe the whole lattice and a determinant computation may miss an index-square factor.

3 Duality, discriminant groups, and class minima

3.1 The dual lattice in coordinates

The dual is

$$L^* = \{y \in V : \langle y, L\rangle \subseteq \mathbb{Z}\}.$$

Writing $y = B\xi$, its pairing column with the lattice basis is $p = G\xi$. Thus

$$y \in L^* \iff p \in \mathbb{Z}^r, \qquad \xi = G^{-1}p, \qquad \|y\|^2 = p^{\mathsf{T}} G^{-1} p.$$

The dual Gram matrix in the dual basis is G^{-1}, and $\det L^* = 1/\det L$. When L is integral, $L \subseteq L^*$. The quotient

$$A_L = L^*/L \cong \mathbb{Z}^r / G\mathbb{Z}^r$$

is the discriminant group. Its order is $\det G$.

More explicitly, if $UGV = \operatorname{diag}(s_1, \dots, s_r)$ is a Smith normal form, where U, V are unimodular and $s_i \mid s_{i+1}$, then

$$A_L \cong \bigoplus_{i=1}^{r} \mathbb{Z}/s_i\mathbb{Z}, \qquad |A_L| = \prod_i s_i = \det G.$$

Smith normal form explains the finite group; the actual verification code can avoid computing it by using fractional-coordinate fingerprints.

3.2 The finite quadratic form and what it does not record

For an even integral lattice, the functions

$$b_L(a, b) = \langle y, z\rangle \bmod \mathbb{Z}, \qquad q_L(a) = \|y\|^2 \bmod 2\mathbb{Z}$$

are well defined for $a = y + L$, $b = z + L$. To verify the second claim, replace y by $y + x$, $x \in L$. The norm changes by $2\langle y, x\rangle + \|x\|^2 \in 2\mathbb{Z}$. Also

$$q_L(a + b) - q_L(a) - q_L(b) = 2b_L(a, b) \pmod{2\mathbb{Z}}.$$

This finite quadratic form controls the parity condition in an even extension. It does not determine the smallest Euclidean norm in a class: the residue of a number modulo $2\mathbb{Z}$ loses its integer part.

Define instead

$$\mu_L(a) = \min_{y \in a} \|y\|^2, \qquad \rho_{\mathrm{glue}}(L)^2 = \max_{a \in A_L} \mu_L(a).$$

The zero class has $\mu_L(0) = 0$, and $\mu_L(-a) = \mu_L(a)$. The number $\rho_{\mathrm{glue}}(L)$ is a *discrete glue radius*. It is the largest distance from a dual lattice point to L. It must not be confused with the ordinary covering radius

$$R(L) = \sup_{v \in V} \operatorname{dist}(v, L), \qquad \rho_{\mathrm{glue}}(L) \le R(L).$$

Equality is not assumed. The exact quantity needed in the proof is the maximum over the finite group A_L.

3.3 Exact fingerprints and a completeness certificate

Let $d = \det G$, $A = \operatorname{adj} G = dG^{-1}$. Both d and A are integral. Associate to a pairing column $p \in \mathbb{Z}^r$ the fingerprint

$$\kappa(p) = Ap \pmod{d} \in (\mathbb{Z}/d\mathbb{Z})^r. \tag{2}$$

Then

$$\kappa(p) = \kappa(q) \iff G^{-1}(p - q) \in \mathbb{Z}^r \iff p - q \in G\mathbb{Z}^r.$$

Consequently fingerprints distinguish precisely the discriminant classes, and there are exactly d possible fingerprints.

Enumerate every integer p satisfying $p^{\mathsf{T}}Ap \le dB$, for a rational bound B, and record the smallest value $p^{\mathsf{T}}Ap/d$ for each fingerprint. If all d fingerprints occur, these recorded numbers are the *exact* class minima. Indeed, each class has a representative of norm at most B, and every vector of smaller norm was included in the exhaustive ball. This is more than finding one short representative per class: the exhaustiveness of the ball is part of the argument.

The program either fixes a bound that covers all classes and verifies the count d, or increases an integral bound until that count is reached. For an arbitrary positive-definite lattice this procedure terminates, because there are finitely many classes, each having a finite minimum. The implementation's emergency bound is a failure guard: reaching it raises an error instead of silently declaring an incomplete table complete.

4 One-vector extensions and the full layer criterion

4.1 The Schur complement as perpendicular squared length

Suppose $J \subset K$ is a saturated corank-one sublattice and G is a Gram matrix of J. Choose $v \in K$ whose image generates K/J. Then a Gram matrix of K is

$$\mathcal{E}(G; p; n) := \begin{pmatrix} G & p \\ p^{\mathsf{T}} & n \end{pmatrix}, \qquad p_i = \langle b_i, v \rangle. \tag{3}$$

Write $v = y + z$, where $y \in \operatorname{span}_{\mathbb{R}} J$, $z \perp J$. Integrality implies $p \in \mathbb{Z}^r$, so $y \in J^*$, with coordinates $G^{-1}p$. Put

$$s = \|z\|^2 = n - p^{\mathsf{T}}G^{-1}p.$$

Block elimination gives

$$\det K = (\det J)s, \qquad K > 0 \iff G > 0 \text{ and } s > 0. \tag{4}$$

Here $K > 0$ denotes positive definiteness. The identity is the algebraic expression of "base volume times height" for a parallelepiped.

Replacing v by $v + x$, $x \in J$, is a unimodular change of basis. It changes y within its class $a \in A_J$, while s is unchanged. We can therefore choose y to be a minimum-norm representative, with $\|y\|^2 = \mu_J(a)$. The last diagonal entry is then $n = s + \mu_J(a)$.

Proposition 4.1 (complete extension criterion)**.** *Assume that J is positive definite, even and integral, and* $\min J \geq 6$. *A choice of $a \in A_J$ and $s \in \mathbb{Q}_{>0}$ defines a positive-definite corank-one lattice extension. It is integral if and only if $s + \mu_J(a) \in \mathbb{Z}$. It is even integral with minimum at least six if and only if*

$$s + q_J(a) = 0 \quad \text{in } \mathbb{Q}/2\mathbb{Z}, \tag{5}$$

$$m^2 s + \mu_J(ma) \geq 6 \quad (m \in \mathbb{Z} \setminus \{0\}). \tag{6}$$

Its determinant is $(\det J)s$.

Proof. Choose $y \in a$ and a new perpendicular vector z of squared length s. Set $K = J + \mathbb{Z}(y+z)$. Pairings between $y+z$ and J are integral. Evenness requires its norm to belong to $2\mathbb{Z}$, which is exactly (5). For any integer m, the layer $J + m(y+z)$ has fixed perpendicular component mz. Its least squared norm is therefore

$$m^2 s + \min_{x \in J} \|x + my\|^2 = m^2 s + \mu_J(ma).$$

For $m = 0$, the zero vector must be omitted when testing the lattice minimum; all remaining vectors already have norm at least six by the hypothesis on J. For $m \neq 0$, every vector is nonzero and (6) is necessary and sufficient. Formula (4) proves the determinant statement. □

The explicit assumption $\min J \geq 6$ is essential. A layer test for $m \neq 0$ cannot repair an already short nonzero vector in the parent. The earlier exposition relied on this hypothesis through the flag context; the proposition above states it in its own right.

4.2 Two finite ways of checking all layers

Because $\mu_J(ma) \geq 0$, only positive integers with

$$m^2 s < 6$$

need testing. Negative m give the same minima. Thus the largest necessary positive layer is $\lceil \sqrt{6/s} \rceil - 1$; a program need not compute this square root and can simply continue while the exact rational inequality $m^2 s < 6$ holds. Equality $m^2 s = 6$ is already safe.

Alternatively, let $h = \operatorname{ord}(a)$. Testing $m = 1, \ldots, h$ is sufficient, since for every $m > h$ its positive residue representative $r \in \{1, \ldots, h\}$ satisfies $ma = ra$ and $r \leq m$. Symmetry allows a shorter test, but it must include the zero residue:

$$\begin{cases} m^2 s + \mu_J(ma) \geq 6, & 1 \leq m \leq \lfloor h/2 \rfloor, \\ h^2 s \geq 6. \end{cases} \tag{7}$$

For each nonzero residue modulo h, choose its representative with smallest absolute value; for the zero residue, the shortest *nonzero* layer index is h, not zero. Omitting the last inequality can falsely accept an extension containing a short perpendicular vector. The audit of one legacy helper found exactly this generic omission; its harmlessness for the actual input is proved in Section 15.

4.3 Why there are only finitely many candidate extensions

Fix an allowed determinant interval $D_- \leq \det K \leq D_+$, put $d = \det J$, and use a shortest representative of class a. Then

$$\mu_J(a) + \frac{D_-}{d} \leq n \leq \mu_J(a) + \frac{D_+}{d}, \qquad n \in 2\mathbb{Z}.$$

There are finitely many possible even values of n, and exactly d possible classes. For every pair (a, n), the Schur complement, parity, determinant congruence, and minimum tests are exact rational or integer tests. This proves finiteness without any unproved bound on the entries of an arbitrary Gram matrix.

Corollary 4.2 (glue-radius lower bound)**.** *Every positive-definite integral one-vector extension $K \supset J$ with minimum at least six satisfies*

$$\det K \geq \det J\big(6 - \rho_{\mathrm{glue}}(J)^2\big). \tag{8}$$

Proof. The layer $m = 1$ gives $s \geq 6 - \mu_J(a)$. Since $\mu_J(a) \leq \rho_{\mathrm{glue}}(J)^2$, multiply by $\det J$. □

This is a necessary bound, not a sufficient construction criterion. Equality additionally requires a deepest discriminant class, correct parity, and all other layer inequalities. If the radius square exceeds six, the bound remains true but need not be useful.

5 Shortest dual vectors and saturated flags

Lemma 5.1 (a primitive dual normal)**.** *Let M have rank $r \geq 2$, and choose a shortest nonzero $y \in M^*$. Then y is primitive in M^*, and $J = M \cap y^\perp$ is a saturated rank-$r - 1$ sublattice. Moreover*

$$\det J = (\det M)\|y\|^2 = (\det M) \min M^*. \tag{9}$$

Proof. If $y = hy_0$, where $h > 1$ is integral and $y_0 \in M^*$, then y_0 is shorter, a contradiction. In the dual basis, the integer coordinates of y therefore have greatest common divisor one. By Bézout's identity the map $x \mapsto \langle x, y\rangle$ from M to $\mathbb{Z}$ is onto. Its kernel is J, and $M/J \cong \mathbb{Z}$, proving rank and saturation.

Choose $v \in M$ with $\langle v, y\rangle = 1$. Its component perpendicular to J is $y/\|y\|^2$, with square norm $1/\|y\|^2$. Apply the Schur complement to the basis (J, v): $\det M = \det J/\|y\|^2$. □

Starting from a hypothetical counterexample L_9, repeatedly apply this lemma to obtain

$$L_1 \subset L_2 \subset \cdots \subset L_8 \subset L_9, \qquad d_r = \det L_r, \tag{10}$$

where every inclusion is saturated, every lattice is even and integral with minimum at least six, and

$$\min L_r^* = d_{r-1}/d_r \qquad (2 \leq r \leq 9).$$

All shortest-vector ties are allowed. No canonical choice of a shortest dual normal is required; at least one such flag exists for every hypothetical counterexample.

For a proposed step $J \subset K$ with Schur complement s, the dual vector orthogonal to J has norm square $1/s$. Therefore the shortest-dual condition is exactly

$$\min K^* = 1/s. \tag{11}$$

Its verification is finite: enumerate every vector of the dual form through the rational bound $1/s$, and check that no nonzero vector has smaller norm. The normal itself occurs, so this test also has an explicit witness at the bound.

6 Hermite bounds and an elementary proof of Mordell's inequality

The Hermite invariant of a rank-r Euclidean lattice is

$$\frac{\min L}{(\det L)^{1/r}}.$$

It is invariant under uniform scaling. The Hermite constant is its supremum over all lattices, without an integrality condition:

$$\gamma_r = \sup_{\operatorname{rk} L = r} \frac{\min L}{(\det L)^{1/r}}.$$

It follows directly that

$$\min L \le \gamma_r (\det L)^{1/r}, \qquad \min L^* \le \gamma_r (\det L)^{-1/r}. \tag{12}$$

The constants are finite. For completeness, a short route to finiteness uses the convex-body lattice theorem: a centrally symmetric convex body of volume greater than $2^r \operatorname{covol} L$ contains a nonzero lattice point. To see the underlying argument, a measurable set of volume greater than one fundamental domain has two distinct points differing by a lattice vector, by translating its pieces into that domain. Apply this to half the symmetric body; its difference set lies in the original body by convexity and symmetry. Taking Euclidean balls gives a finite upper bound for γ_r. No numerical version of that bound enters the search.

Theorem 6.1 (classical low-dimensional Hermite theorem; external input)**.** *For $1 \le r \le 8$, the exact values are*

r	1	2	3	4	5	6	7	8
γ_r^r	1	4/3	2	4	8	64/3	64	256.

In dimension eight, equality in the Hermite bound is attained only by lattices similar to E_8.

This is a substantive external theorem, not a fact proved by the finite programs. The values are tabulated, for example, by Wen and Chang [6]; the classical lattice sources are [3, 4, 5]. The dimension-eight equality statement is also a consequence of Viazovska's optimality and uniqueness theorem for periodic packings [7], because a lattice packing is periodic. We state this dependency explicitly instead of silently turning an enumeration into a proof of the Hermite constants themselves.

Proposition 6.2 (Mordell's inequality)**.** *For $r \ge 3$,*

$$\gamma_r \le \gamma_{r-1}^{(r-1)/(r-2)}. \tag{13}$$

Proof. Scale any rank-r lattice M to determinant one. Choose a shortest nonzero vector of M^*, and let J be its primitive orthogonal section. Lemma 5.1 gives $\det J = \min M^* \le \gamma_r$. Since $J \subset M$,

$$\min M \le \min J \le \gamma_{r-1} (\det J)^{1/(r-1)} \le \gamma_{r-1} \gamma_r^{1/(r-1)}.$$

Take the supremum over all determinant-one M. Finiteness and positivity of γ_r justify division and give $\gamma_r^{(r-2)/(r-1)} \le \gamma_{r-1}$, which is the claim. No existence of a lattice attaining the supremum is needed. □

For the rank-nine proof this gives

$$\gamma_9 \le \gamma_8^{8/7} = 2^{8/7}.$$

Applying (12) to each stage of (10) yields the two complementary bounds

$$d_r \geq 6^r/\gamma_r^r, \tag{14}$$

$$d_{r-1} \leq \gamma_r d_r^{(r-1)/r}. \tag{15}$$

The first prevents a lattice of minimum six from having too small a determinant. The second prevents its shortest-dual section from having too large a determinant. Their interaction makes a search beginning in rank one possible.

7 Determinant congruences and the exact flag windows

7.1 The congruence modulo four

Lemma 7.1. *If G is a symmetric integral $r \times r$ matrix with even diagonal, then its determinant is even for odd r. If $r = 2m$,*

$$\det G \equiv 0 \text{ or } (-1)^m \pmod 4.$$

Proof. Over $\mathbb{F}_2$, the form defined by G is alternating, since its diagonal vanishes. An alternating form has even rank: if it is nonzero, choose vectors with pairing one, split off their nonsingular two-dimensional plane, and continue on the orthogonal complement. Thus in odd dimension the determinant is zero modulo two.

Now let $r = 2m$. If $G \bmod 2$ is singular, its nullity is an even positive number, hence at least two. At least two of its Smith invariant factors are even, so $4 \mid \det G$. Otherwise an odd off-diagonal entry gives an invertible block over $\mathbb{Z}/4\mathbb{Z}$ of the form

$$P = \begin{pmatrix} 2a & b \\ b & 2c \end{pmatrix}, \qquad b \text{ odd}.$$

Its determinant is $4ac - b^2 \equiv -1 \pmod 4$. Orthogonal block elimination over $\mathbb{Z}/4\mathbb{Z}$ splits off this block: invertibility of P allows the cross block to be killed by a triangular congruence matrix. Congruence preserves the property of having even diagonal. The remaining block is nonsingular modulo two and the process repeats. There are m blocks. A change of basis multiplies the determinant by the square of a unit modulo four, which equals one, so the product is $(-1)^m \pmod 4$. □

In particular, a hypothetical rank-nine determinant strictly below 21870 satisfies

$$d_9 \leq 21868. \tag{16}$$

The even-rank restrictions alternate: ranks $2, 6$ allow residues $0, 3$, while ranks $4, 8$ allow residues $0, 1$.

7.2 The exceptional rank-eight equality case

The Hermite lower bound gives $d_8 \geq 6^8/256 = 6561 = 3^8$. Equality forces $L_8 \cong \sqrt{3}E_8$ by Theorem 6.1. We exclude it within a hypothetical counterexample before raising the lower end of the rank-eight window.

A standard model of E_8 is

$$D_8 \ \cup \ \big((\tfrac{1}{2}, \ldots, \tfrac{1}{2}) + D_8\big), \qquad D_8 = \{x \in \mathbb{Z}^8 : \textstyle\sum x_i \text{ is even}\}.$$

This is a lattice because twice the half-vector belongs to D_8. It is integral and even; a direct calculation verifies the pairings and parity in both cosets. Since D_8 has index two in $\mathbb{Z}^8$, it has

determinant four. The displayed union has index two over D_8, so E_8 has determinant one by (1). Its minimum is two: nonzero integral vectors of even coordinate sum have norm at least two, half-integral vectors have norm at least $8/4 = 2$, and this bound is attained. Thus E_8 is even unimodular and $E_8^* = E_8$.

For $J = \sqrt{3}E_8$, dual classes can be written as

$$J^*/J = (E_8/\sqrt{3})/(\sqrt{3}E_8) \cong E_8/3E_8.$$

There are $3^8 = 6561$ classes. Exact enumeration gives the following distribution of minimum *unscaled* norms in $E_8/3E_8$:

minimum norm in the residue class	0	2	4	6	8
number of classes	1	240	2160	2240	1920

The counts add to 6561. Exhaustive enumeration through norm eight, together with occurrence of every class, makes the minima exact by the argument in Section 3. Scaling the vectors by $1/\sqrt{3}$ gives

$$\rho_{\mathrm{glue}}(\sqrt{3}E_8)^2 = 8/3.$$

Consequently a rank-nine continuation has determinant at least

$$6561(6 - 8/3) = 6561 \cdot 10/3 = 21870.$$

This contradicts (16). The next integer above 6561 allowed by Lemma 7.1 is 6564.

7.3 Integer-power verification of every endpoint

The top upper bound is

$$d_8 \le 2^{8/7} d_9^{8/9} \le 2^{8/7} 21868^{8/9}.$$

Its floor is certified without floating point by

$$15908^{63} \le 2^{72} 21868^{56} < 15909^{63}. \tag{17}$$

For subsequent stages let $c_r = \gamma_r^r = a_r/b_r$ in reduced form. A nonnegative integer u satisfies $u \le \gamma_r D^{(r-1)/r}$ precisely when

$$b_r u^r \le a_r D^{r-1}.$$

Thus the raw floor U is certified by a pair of integer inequalities

$$b_r U^r \le a_r D^{r-1} < b_r (U+1)^r. \tag{18}$$

Round this floor down further only if the determinant congruence requires it. The complete recursion is:

parent rank	parent upper bound	raw child floor	congruence-adjusted floor
9	21868	15908	15908
8	15908	9493	9492
7	9492	4647	4647
6	4647	1894	1894
5	1894	634	633
4	633	178	178
3	178	39	39
2	39	7	6

All raw floors are checked by (17) or (18). Combining them with the lower Hermite bounds and the excluded E_8 case yields

r	determinant window	congruence
8	$6564 \le d_8 \le 15908$	$d_8 \equiv 0, 1 \pmod 4$
7	$4374 \le d_7 \le 9492$	$d_7 \equiv 0 \pmod 2$
6	$2187 \le d_6 \le 4647$	$d_6 \equiv 0, 3 \pmod 4$
5	$972 \le d_5 \le 1894$	$d_5 \equiv 0 \pmod 2$
4	$324 \le d_4 \le 633$	$d_4 \equiv 0, 1 \pmod 4$
3	$108 \le d_3 \le 178$	$d_3 \equiv 0 \pmod 2$
2	$27 \le d_2 \le 39$	$d_2 \equiv 0, 3 \pmod 4$
1	$d_1 = 6$	$d_1 \equiv 0 \pmod 2$.

(19)

For example, the lower rank-seven endpoint is $6^7/64 = 4374$, and the lower rank-six endpoint is $6^6/(64/3) = 2187$. The rank-one window forces the unique seed Gram matrix [6].

7.4 A transparent rank-two example

For the seed [6], dual classes are indexed by $j \bmod 6$, and the shortest representatives have pairing columns $j = 0, \pm 1, \pm 2, 3$, with class minima $j^2/6$. Bordering the parent gives

$$\begin{pmatrix} 6 & j \\ j & n \end{pmatrix}, \qquad \det = 6n - j^2.$$

The determinant interval $[27, 39]$, evenness, and the exact flag tests leave seven labelled presentations. The two signs for $j = 1$ and for $j = 2$ each give an isometric pair. The five representatives may be written

$$\begin{pmatrix} 6 & 3 \\ 3 & 6 \end{pmatrix}, \quad \begin{pmatrix} 6 & 2 \\ 2 & 6 \end{pmatrix}, \quad \begin{pmatrix} 6 & 1 \\ 1 & 6 \end{pmatrix}, \quad \begin{pmatrix} 6 & 0 \\ 0 & 6 \end{pmatrix}, \quad \begin{pmatrix} 6 & 3 \\ 3 & 8 \end{pmatrix}.$$

Their determinants are $27, 32, 35, 36, 39$. This small example illustrates both exhaustiveness by discriminant classes and why a labelled child count differs from an isometry-class count.

8 The dual-vector lifting inequality

Lemma 8.1. *Let $J \subset K$ be a saturated corank-one inclusion for which the normal to J is shortest in K^*. If $s = \det K / \det J$ and $\lambda = \min J^*$, then*

$$s \ge \frac{3}{4\lambda}.$$

Proof. Write $K = J + \mathbb{Z}v$, and let $y_0 \in K^*$ be the perpendicular normal with $\langle y_0, v \rangle = 1$. Then $\|y_0\|^2 = 1/s$. Choose $u \in J^*$ with $\|u\|^2 = \lambda$. Rounding $\langle u, v \rangle$ to a nearest integer gives $c \in [-1/2, 1/2]$ with $\langle u, v \rangle + c \in \mathbb{Z}$. Hence $u + cy_0 \in K^*$. This vector is nonzero because its projection u to J's span is nonzero. Orthogonality gives

$$\|u + cy_0\|^2 = \lambda + c^2/s \le \lambda + 1/(4s).$$

Since y_0 is shortest in K^*, $1/s \le \lambda + 1/(4s)$. Rearranging proves the claim. □

The constant $1/4$ comes from the squared maximum distance to the nearest integer. It is a rigorous universal rounding bound; no probabilistic argument is involved. In the consecutive flag steps $L_6 \subset L_7 \subset L_8$, the previous normal identity gives $\min L_7^* = d_6/d_7$. Thus

$$d_8 \ge \frac{3d_7^2}{4d_6}. \tag{20}$$

Any determinant pair for which the right side exceeds 15908 can be discarded before enumerating rank-eight extensions. Equality at the upper endpoint would not permit discarding the pair.

The same inequality is used as a parent-level optimization in the bottom-up generator: if $D_+/\det J < 3/(4\min J^*)$, no child in the window can have its normal shortest. The generator returns an empty list together with the exact inequality, rather than relying on a heuristic estimate for the dual minimum.

9 How the exact finite computation works

9.1 Rational elimination and positive definiteness

Every Gram matrix and pairing vector is integral. Inverses, Schur complements, class minima, and recursion bounds are represented by reduced rational numbers. An exact LDL^{T} decomposition writes

$$A = L\,\mathrm{diag}(d_1,\dots,d_r)L^{\mathsf{T}},$$

where L is unit lower triangular. Recursively,

$$d_k = A_{kk} - \sum_{h<k} L_{kh}^2 d_h,$$
$$L_{ik} = \frac{A_{ik} - \sum_{h<k} L_{ih}L_{kh}d_h}{d_k} \quad (i>k).$$

All $d_k > 0$ certify positive definiteness, since this is a congruence to a positive diagonal matrix. Reconstruction of A can be checked entry by entry. The same formulas apply to a positive-definite rational matrix such as G^{-1}.

Bareiss elimination computes integer determinants while controlling denominator growth. If the current pivot is p and the previous pivot is p_{old}, the trailing entries are updated by

$$a_{ij} \longleftarrow \frac{p\,a_{ij} - a_{ik}a_{kj}}{p_{\mathrm{old}}}.$$

The initial previous pivot is one; the divisions are exact by the determinantal identities underlying fraction-free elimination. Row swaps are accompanied by a sign change. An exact implementation should verify divisibility instead of treating integer truncation as a proof. In the audited positive-definite Gram matrices all leading pivots are nonzero. Unimodular isometry matrices need row pivoting and an appropriate sign check. The supplementary audit also compared determinants with an independent permutation formula on small cases and on the displayed rank-nine matrix.

9.2 Exact ellipsoid enumeration and integer endpoints

The decomposition gives

$$x^{\mathsf{T}}Ax = \sum_{k=1}^{r} d_k \left(x_k + \sum_{j>k} L_{jk}x_j \right)^2. \tag{21}$$

Enumerate coordinates from x_r down to x_1. Suppose the higher coordinates are fixed and have used norm $U \le B$. At level k, put

$$c = -\sum_{j>k} L_{jk}x_j, \qquad R = (B-U)/d_k.$$

The allowed coordinates satisfy $(x_k - c)^2 \le R$. Write $c = a/b$, $b > 0$, and $R = p/q \ge 0$, $q > 0$. Then

$$(bx_k - a)^2 \le b^2 p/q.$$

The left side is an integer square. Define

$$N = \left\lfloor \sqrt{\lfloor b^2 p/q \rfloor} \right\rfloor .$$

The exact coordinate interval is

$$\left\lceil \frac{a - N}{b} \right\rceil \le x_k \le \left\lfloor \frac{a + N}{b} \right\rfloor . \tag{22}$$

All operations use integer division and integer square root. Flooring before taking the integer square root is exact here because the quantity being bounded is itself an integer square. Another audited enumerator uses a slightly larger interval from a rational square-root ceiling and then an exact membership test; that method is also complete.

To prove exhaustiveness, take any x with $x^{\mathsf{T}} A x \le B$. Each partial sum in (21) is nonnegative and at most B. At each recursion level its coordinate therefore lies in (22). Induction down the coordinates shows that the algorithm visits x. Conversely, every emitted vector satisfies the exact sum bound. This proves both directions, including vectors exactly on the boundary. Floating-point rounding cannot delete a boundary vector, because it is never used in a membership decision.

9.3 Isometries and the safe role of invariants

To test whether A is isometric to R, an exhaustive method enumerates all target vectors of the prescribed basis norms A_{ii}, then chooses columns x_i satisfying

$$x_i^{\mathsf{T}} R x_j = A_{ij} \quad \text{for all } i, j.$$

Each pool is finite and can be generated by the exact ellipsoid routine. At a completed assignment, verify $|\det X| = 1$. This is a complete search for a basis isometry.

Short-vector shell counts and pairing histograms are useful filters: different invariants can rule out isometry, while equal invariants only trigger the full test. The shell bound must be common to the forms being compared; a basis-dependent cutoff is not an isometry invariant. Most importantly, every *positive merger* is certified by an explicit unimodular matrix. A failure to merge two isometric forms only increases work. An unsupported merger can delete a necessary branch and would invalidate the proof. The archived maps make all positive mergers directly checkable independently of the search heuristic used to find them.

9.4 The induction proving completeness after deduplication

Let $\mathcal{R}_r$ be the retained representatives in rank r. The induction invariant is:

> Every rank-r lattice occurring in a hypothetical flag satisfying (19) is isometric to a representative in $\mathcal{R}_r$.

For $r = 1$, the window forces [6], so the invariant holds. Assume it holds for rank r, and consider an actual flag inclusion $L_r \subset L_{r+1}$. Transfer it along an isometry of L_r to its retained representative J. Saturation writes the child as a one-vector extension of J. Translation of its generator by J makes its projection a shortest representative of a discriminant class. That class occurs in the complete class table, and its even last diagonal entry lies in the enumerated interval. The layer and shortest-normal tests accept the transferred child, because those are isometry-invariant conditions that the actual flag satisfies. Deduplication then sends it to a retained representative by an explicit integral isometry. This proves the invariant in rank $r + 1$.

Deduplication need not remember the entire earlier flag. An isometry of L_r transports that flag, so the necessary existence of an earlier flag is preserved. The later extension tests

use only the current parent lattice and its actual dual minimum; they do not require a fixed basis embedding of L_{r-1} into the representative. At the rank-seven pruning stage the recorded determinant pair supplies $d_6/d_7 = \min L_7^*$, an invariant of L_7. This explains why forgetting earlier labelled embeddings cannot remove a valid counterexample.

10 Enumeration data and reduction to six branches

The lower flag starts with [6]. Complete regeneration gives

child rank	labelled children	retained isometry classes
2	7	5
3	52	14
4	279	47
5	1456	118
6	1905	138

The total number of labelled children in these five stages is $7 + 52 + 279 + 1456 + 1905 = 3699$. Each labelled child has a checked unimodular map to its retained representative. Additional bridge certificates identify the regenerated representative sets with the frozen sets used by the later computations. These bridges are needed because two equally valid runs can select different bases or representatives.

All 138 rank-six parents are then extended. Four disjoint index ranges cover $[0, 138)$; their child counts are

$$102, \quad 0, \quad 1868, \quad 265, \qquad 102 + 0 + 1868 + 265 = 2235.$$

The zero-count chunk is included and checked: it is not a missing range. Ten parents have children. The 2235 labelled forms realize 31 distinct pairs (d_6, d_7) and 28 distinct rank-seven determinants. The boundary dual enumeration in their validation contains 5222 vectors; boundary vectors are retained so that shortest-vector ties are tested correctly.

Applying (20) eliminates 25 pairs and 2139 forms. The six remaining pairs contain 96 forms:

d_6	d_7	labelled count	representative
2187	4374	2	$\sqrt{3}E_7$
2187	6318	54	H_{6318}
2835	7722	32	H_{7722}
4096	8192	1	$2E_7^*$
3240	8100	1	H_{8100}
3267	7986	6	H_{7986}

(23)

The sum is $2 + 54 + 32 + 1 + 1 + 6 = 96$, and $2139 + 96 = 2235$. Every survivor is linked by an explicit $\mathrm{GL}_7(\mathbb{Z})$-map to the representative in its row.

These are six representatives sufficient for the counterexample search. They are not asserted to constitute all even integral rank-seven lattices of minimum six. The determinant windows and the existence of the selected shortest-dual flag impose additional restrictions.

11 Explicit matrices and closure of all six branches

When the parent is fixed, write $\mathcal{E}(G; p; n)$ for the block matrix (3). Giving G, p, n specifies every entry of the extension matrix and is easier to inspect than repeating a large common block. The following choices agree with the certificate representatives, or are explicitly linked to them by the archived integral isometries.

11.1 The root-lattice branch $\sqrt{3}E_7$

Use

$$H = \begin{pmatrix} 6 & -3 & -3 & -3 & -3 & -3 & -3 \\ -3 & 6 & 3 & 3 & 3 & 3 & 3 \\ -3 & 3 & 6 & 0 & 3 & 0 & 0 \\ -3 & 3 & 0 & 6 & 3 & 3 & 3 \\ -3 & 3 & 3 & 3 & 6 & 3 & 3 \\ -3 & 3 & 0 & 3 & 3 & 6 & 3 \\ -3 & 3 & 0 & 3 & 3 & 3 & 6 \end{pmatrix}. \tag{24}$$

It has determinant $4374 = 2 \cdot 3^7$ and minimum six. The matrix $H/3$ is a Gram matrix of a root basis of E_7. Because some off-diagonal entries are positive, it is not being called a Cartan matrix of a simple-root basis.

Its 4374 discriminant classes have minimum distribution

minimum	0	1/2	2/3	7/6	4/3	11/6	2	5/2	8/3	19/6	9/2
count	1	56	126	576	756	756	728	672	576	126	1

Thus $\rho_{\mathrm{glue}}(H)^2 = 9/2$. The broad extension enumeration in the rank-eight window accepts 1375 labelled discriminant classes: 126 have determinant 12393, 576 have determinant 14580, and 673 have determinant 15309. Root reflections act on the finite group and group these into four extension types.

For clarity, a reflection in a norm-two root b_i of the root lattice is $x \mapsto x - \langle x, b_i\rangle b_i$. If $C = H/3$ and p is a pairing column for the scaled lattice, the induced pairing update is

$$p_j \longmapsto p_j - p_i C_{ij}.$$

It is an integral isometry and preserves class minima. The computation forms the orbits of these explicitly verified maps, together with any stated sign action. Completeness does not require an unproved assertion that these generators give the full Weyl group: their subgroup orbits already cover the finite class set, and the listed orbit-to-target identifications suffice.

The four target matrices are

$$\begin{aligned} J_{12393} &= \mathcal{E}\big(H; (1,-3,0,-3,-3,-3,-3)^{\mathsf{T}}; 6\big), \\ J_{14580} &= \mathcal{E}\big(H; (2,-3,-1,-3,-3,-3,-3)^{\mathsf{T}}; 6\big), \\ J_{15309,a} &= \mathcal{E}\big(H; (0,2,2,-1,-1,-1,-1)^{\mathsf{T}}; 6\big), \\ J_{15309,b} &= \mathcal{E}\big(H; (0,-3,0,-3,-3,-3,-3)^{\mathsf{T}}; 8\big). \end{aligned}$$

The two forms of determinant 15309 are distinct targets; determinant alone is not used to identify lattices. Their exact class tables give

target	dual vectors enumerated	ρ^2_{glue}	rank-nine determinant floor
J_{12393}	73253	58/17	$12393 \cdot 44/17 = 32076$
J_{14580}	190491	9/2	$14580 \cdot 3/2 = 21870$
$J_{15309,a}$	314953	32/7	$15309 \cdot 10/7 = 21870$
$J_{15309,b}$	72119	68/21	$15309 \cdot 58/21 = 42282$

Every table covers exactly the number of discriminant classes equal to the target determinant. The vector counts are reproducibility diagnostics, not substitutes for those completeness checks. The least rank-nine floor in this branch is 21870.

11.2 The branch H_{6318}

Let G_6 be the leading 6×6 block of H. Then G_6 is a Gram matrix of $\sqrt{3}E_6$, with determinant 2187, and set

$$H_{6318} = \mathcal{E}\big(G_6; (2,-3,-2,-3,-3,-3)^{\mathsf{T}}; 6\big)\,.$$

The exact class table contains all 6318 classes and has $\rho_{\mathrm{glue}}(H_{6318})^2 = 105/26$. A broad enumeration, retaining all minimum-six extensions in the rank-eight window, gives twelve labelled children: two isometric to J_{12393} and ten isometric to

$$J_{13689} = \mathcal{E}\big(H_{6318}; (1,2,2,0,0,0,-3)^{\mathsf{T}}; 6\big)\,.$$

Every positive identification has an explicitly checked unimodular map. The exact radius of this new target is $10/3$, so

$$\det L_9 \geq 13689(6 - 10/3) = 13689 \cdot 8/3 = 36504.$$

For J_{12393}, the already computed floor is 32076.

There is a useful distinction between unrestricted children and actual flag continuations. For this parent pair,

$$\frac{3 \cdot 6318^2}{4 \cdot 2187} = 13689.$$

Thus the determinant-12393 children cannot be steps in the selected shortest-dual flag. Keeping them in the broad enumeration is safe: it adds possibilities, and those added possibilities close as well.

11.3 The branch H_{7722}

A representative is

$$H_{7722} = \begin{pmatrix} 6 & -3 & -3 & -3 & -3 & -3 & 2 \\ -3 & 6 & 3 & 1 & 3 & 3 & -3 \\ -3 & 3 & 6 & 0 & 3 & 0 & -2 \\ -3 & 1 & 0 & 6 & 3 & 3 & -3 \\ -3 & 3 & 3 & 3 & 6 & 3 & -3 \\ -3 & 3 & 0 & 3 & 3 & 6 & -3 \\ 2 & -3 & -2 & -3 & -3 & -3 & 6 \end{pmatrix}.$$

The exact dual minimum is $105/286 = 2835/7722$, as required by its flag pair. Its glue-radius square is $93/22$, so every unrestricted extension has determinant at least

$$7722(6 - 93/22) = 7722 \cdot 39/22 = 13689.$$

The unrestricted enumeration yields exactly two opposite classes, each giving determinant 13689. One presentation is

$$\mathcal{E}\big(H_{7722}; (1,0,2,0,0,0,-3)^{\mathsf{T}}; 6\big)\,.$$

It is explicitly integrally isometric to J_{13689}.

However, the shortest-dual flag condition gives the stronger bound

$$d_8 \geq \frac{3 \cdot 7722^2}{4 \cdot 2835} = \frac{552123}{35} > 13689. \tag{25}$$

The decimal approximation $15774.942857\ldots$ is only explanatory; the exact rational inequality is used. Hence neither of those two unrestricted children is flag-compatible. This branch has *no* rank-eight continuation in the hypothetical flag. Calling the determinant-13689 forms admissible flag continuations would be incorrect. As a redundant check, even these additional forms have rank-nine floor 36504.

For opposite classes, replacing the extension vector by its negative gives a sign isometry if the representative projections are exact negatives. If chosen shortest representatives differ by a lattice vector, first translate the generator by that vector. The resulting matrix is unimodular. The certificate checks the actual 8×8 matrix and its determinant -1, rather than relying on a label “opposite”.

11.4 The determinant-8192 branch and the scaling correction

Use the actual simple-root Cartan matrix

$$C = \begin{pmatrix} 2 & 0 & -1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & -1 & 0 & 0 & 0 \\ -1 & 0 & 2 & -1 & 0 & 0 & 0 \\ 0 & -1 & -1 & 2 & -1 & 0 & 0 \\ 0 & 0 & 0 & -1 & 2 & -1 & 0 \\ 0 & 0 & 0 & 0 & -1 & 2 & -1 \\ 0 & 0 & 0 & 0 & 0 & -1 & 2 \end{pmatrix}.$$

It has determinant two. The target Gram matrix is $4C^{-1}$, and therefore

$$\det(4C^{-1}) = 4^7/2 = 8192.$$

Since multiplying vectors by two multiplies their Gram matrix by four, the lattice is $2E_7^*$. The earlier notation $4E_7^*$ was a scaling error in the prose; the matrix $4C^{-1}$ in the computation has always been the determinant-8192 object.

Its exact radius square is $7/2$, from all 8192 discriminant classes. Thus

$$d_8 \geq 8192(6 - 7/2) = 8192 \cdot 5/2 = 20480 > 15908.$$

No rank-eight child in the window is possible.

11.5 The branch H_{8100}

The chosen matrix is

$$H_{8100} = \begin{pmatrix} 6 & 3 & 0 & -3 & -3 & -3 & -3 \\ 3 & 6 & -3 & 0 & -3 & -3 & -3 \\ 0 & -3 & 6 & -3 & 0 & 0 & 0 \\ -3 & 0 & -3 & 6 & 0 & 2 & 2 \\ -3 & -3 & 0 & 0 & 6 & 1 & 1 \\ -3 & -3 & 0 & 2 & 1 & 6 & 1 \\ -3 & -3 & 0 & 2 & 1 & 1 & 6 \end{pmatrix}.$$

Its exact radius square is $15/4$, certified by 8100 classes. The continuation bound is

$$d_8 \geq 8100(6 - 15/4) = 8100 \cdot 9/4 = 18225 > 15908.$$

Again no rank-eight child in the required window is possible.

11.6 The branch H_{7986}

Use

$$H_{7986} = \begin{pmatrix} 6 & 3 & 0 & -3 & -3 & -3 & -3 \\ 3 & 6 & -3 & 0 & -3 & -3 & -2 \\ 0 & -3 & 6 & -3 & 0 & 1 & -1 \\ -3 & 0 & -3 & 6 & 1 & 2 & 3 \\ -3 & -3 & 0 & 1 & 6 & 0 & 0 \\ -3 & -3 & 1 & 2 & 0 & 6 & 2 \\ -3 & -2 & -1 & 3 & 0 & 2 & 6 \end{pmatrix}.$$

The class table gives $\rho_{\mathrm{glue}}(H_{7986})^2 = 25/6$. Consequently

$$d_8 \geq 7986(6 - 25/6) = 7986 \cdot 11/6 = 14641.$$

The exact flag extension test finds two opposite presentations, both isometric to

$$J_{14641} = \mathcal{E}\big(H_{7986}; (-2,-3,0,2,0,2,2)^{\mathsf{T}}; 6\big)\,.$$

Here the normal is shortest: these are actual flag-compatible extensions, unlike the determinant-13689 children of H_{7722}. The rank-eight target has exactly 14641 discriminant classes and $\rho_{\mathrm{glue}}(J_{14641})^2 = 36/11$. Therefore

$$d_9 \geq 14641(6 - 36/11) = 14641 \cdot 30/11 = 39930.$$

11.7 The six branches in one logical table

rank-seven parent	what happens at rank eight	rank-nine floor
$\sqrt{3}E_7$	four listed unrestricted types	≥ 21870
H_{6318}	two unrestricted types; both closed	≥ 32076
H_{7722}	no flag-compatible continuation	—
$2E_7^*$	rank-eight floor $20480 > 15908$	—
H_{8100}	rank-eight floor $18225 > 15908$	—
H_{7986}	one flag-compatible type, determinant 14641	≥ 39930

Each row uses either a direct obstruction in rank eight or an exact rank-nine determinant floor. Every surviving possibility is covered, and none allows a determinant below 21870.

12 Assembly of the lower-bound proof

Proof of the lower-bound part of Theorem 1.2. Suppose that L is a positive-definite even integral rank-nine lattice of minimum at least six and determinant below 21870. By Lemma 7.1, its determinant is at most 21868. Repeated shortest-dual sections produce the saturated flag (10). The Hermite bounds and Mordell's inequality give the upper and lower windows. The equality case $d_8 = 6561$ closes separately by the exact $\sqrt{3}E_8$ class table. Thus every stage has determinant in (19).

The complete extension criterion and the induction in Section 9.4 show that the flag is represented in the regenerated search. It reaches one of 2235 labelled rank-seven forms. The lifting inequality excludes 2139, and explicit unimodular maps identify all 96 survivors with the six representatives in (23). Section 11 excludes every continuation compatible with the flag and the determinant windows. This contradicts the assumed existence of L, proving $\det L \geq 21870$. □

The finite computations enter at identifiable points: the lower flag lists, their completeness and isometry maps, the rank-seven list and its linkage, and the exact class minima and extension lists in the branches. The implication from these finite facts to the universal theorem is the ordinary mathematical argument above. A transcript containing the expected counts alone would not establish those finite facts.

13 The equality witness and a small independent sharpness proof

The source manuscript provides the symmetric matrix

$$K_9 = \begin{pmatrix} 6 & -3 & -3 & -3 & -3 & -3 & -3 & 3 & -2 \\ -3 & 6 & 3 & 3 & 3 & 3 & 3 & 0 & 1 \\ -3 & 3 & 6 & 0 & 3 & 0 & 0 & -3 & 3 \\ -3 & 3 & 0 & 6 & 3 & 3 & 3 & 0 & -1 \\ -3 & 3 & 3 & 3 & 6 & 3 & 3 & -3 & 1 \\ -3 & 3 & 0 & 3 & 3 & 6 & 3 & 0 & 0 \\ -3 & 3 & 0 & 3 & 3 & 3 & 6 & 0 & 1 \\ 3 & 0 & -3 & 0 & -3 & 0 & 0 & 6 & -1 \\ -2 & 1 & 3 & -1 & 1 & 0 & 1 & -1 & 6 \end{pmatrix}. \tag{26}$$

It is integral and even by inspection. The archived verifier checks positive definiteness, determinant 21870, absence of nonzero vectors of norm below six, and the presence of 258 oriented minimum vectors. These checks were rerun in the current audit.

There is also a much smaller argument for its minimum, reducing the remaining check to 64 explicitly bounded integer tuples. This argument was reconstructed during the audit and does not rely on the archived dual-ellipsoid enumerator.

Let G be the leading 8×8 block of K_9, set $C = G/3$, and let

$$p = (-2, 1, 3, -1, 1, 0, 1, -1)^{\mathsf{T}}.$$

Exact elementary matrix calculations give

$$\det C = 1, \qquad C > 0, \qquad \operatorname{diag}(C^{-1}) = (4, 2, 4, 2, 4, 2, 2, 2), \tag{27}$$

$$w = C^{-1}p = (-1, -1, 2, -1, 0, 0, 1, 1)^{\mathsf{T}}, \qquad w^{\mathsf{T}}Cw = 8. \tag{28}$$

For instance $C > 0$ can be checked by its positive rational LDL^{T} pivots, a short matrix calculation. It has even diagonal and is integral, so every nonzero integer vector has norm at least two, and a basis vector attains two. Thus $G = 3C$ has minimum six and determinant $3^8 = 6561$.

The projection of the ninth basis vector has coordinates $G^{-1}p = w/3$, so its squared norm is

$$p^{\mathsf{T}}G^{-1}p = 8/3.$$

Its Schur complement is

$$s = 6 - 8/3 = 10/3. \tag{29}$$

This proves positive definiteness of K_9 and

$$\det K_9 = 6561 \cdot 10/3 = 21870.$$

To prove the minimum by the layer criterion, it remains only to show that the projected discriminant class has minimum exactly $8/3$.

After multiplying coordinates by three, this amounts to showing

$$\min_{r \in w + 3\mathbb{Z}^8} r^{\mathsf{T}}Cr = 8. \tag{30}$$

For $r = w + 3x$, even integrality gives

$$r^{\mathsf{T}}Cr - w^{\mathsf{T}}Cw = 6x^{\mathsf{T}}Cw + 9x^{\mathsf{T}}Cx \in 6\mathbb{Z}.$$

Thus a norm smaller than eight in this residue class could only be two. For a vector of norm two, Cauchy–Schwarz in the inner product defined by C implies

$$|r_i|^2 = |\langle C^{-1}e_i, r\rangle_C|^2 \le (C^{-1})_{ii}\, r^{\mathsf T} C r \le 8.$$

All coordinates are therefore integers of absolute value at most two. Combining these bounds with $r \equiv w \pmod 3$ leaves

$$r_1, r_2, r_3, r_4 \in \{-1, 2\}, \quad r_5 = r_6 = 0, \quad r_7, r_8 \in \{-2, 1\}.$$

There are $2^6 = 64$ tuples. Direct evaluation of $r^{\mathsf T} C r$, using integer additions and multiplications, gives:

norm	8	14	20	26	32	38	44	50	56	62	86
count	5	9	14	10	8	5	4	4	1	3	1

The counts add to 64, and norm two does not occur. Since w itself has norm eight, (30) follows. A short direct reproduction appears in Appendix A.

We have now proved $\mu_G(a) = 8/3$. The layers $m = \pm 1$ have minimum

$$10/3 + 8/3 = 6.$$

Every layer with $|m| \ge 2$ has perpendicular norm at least $4 \cdot 10/3 = 40/3 > 6$. The parent layer already has minimum six. Hence $\min K_9 = 6$. This proves the upper-bound part of Theorem 1.2 and explains structurally why the threshold 21870 appears: it is $\det(\sqrt{3}E_8)$ times the least perpendicular squared length permitted by a deepest dual class.

The auxiliary count 258 is a useful independent diagnostic but is not required to establish the minimum or the determinant optimum. The 64-tuple argument is not claimed to enumerate all vectors of norm eight; its coordinate bounds were derived for a hypothetical norm-two vector and are used only to rule that possibility out.

14 Orthogonal complements in unimodular lattices and packing density

14.1 The determinant identity for a primitive section

Let Λ be integral unimodular, meaning $\det \Lambda = 1$, equivalently $\Lambda^* = \Lambda$. Let $K \subset \Lambda$ be saturated, and define

$$L = \Lambda \cap (\operatorname{span}_{\mathbb R} K)^{\perp}.$$

Then L is saturated as well: if nx is orthogonal to K for a nonzero integer n, so is x. We prove

$$\det L = \det K. \tag{31}$$

Because K is primitive as an abelian subgroup, a basis of it extends to a basis of Λ. Any integer-valued linear functional on K therefore extends to one on Λ. Unimodularity identifies such functionals with pairings against elements of Λ. Consequently the restriction map

$$\Lambda \longrightarrow K^*, \qquad x \longmapsto \big(k \mapsto \langle x, k\rangle\big)$$

is onto, and its kernel is L. Modding out the image of K gives

$$\Lambda/(K \oplus L) \cong K^*/K.$$

Put $h = [\Lambda : K \oplus L]$. Then $h = \det K$. By orthogonality and the index-square formula,

$$\det K \det L = \det(K \oplus L) = h^2 \det \Lambda = h^2.$$

Substituting $h = \det K$ proves (31).

This derivation also pinpoints why unimodularity and primitivity cannot be suppressed from the statement. For a nonsaturated section, the quotient has extra torsion; for a nonunimodular ambient lattice, the restriction map need not be onto in the same way.

14.2 What density the determinant controls

Suppose Λ has rank 48, is even unimodular, and has minimum six. A primitive rank-nine section K and its orthogonal complement L have ranks nine and 39. Both inherit minimum at least six. Put balls of radius

$$r_0 = \sqrt{6}/2 = \sqrt{3/2}$$

around points of L. Their interiors are disjoint. The center density, which omits the volume of the unit ball, is

$$\delta_{\mathrm{bare}}(L) = \frac{r_0^{39}}{\operatorname{covol} L} = \frac{(3/2)^{39/2}}{\sqrt{\det K}}.$$

Ordinary packing density is this quantity multiplied by $\mathrm{vol}(B_{39}(1))$. Theorem 1.2 implies

$$\delta_{\mathrm{bare}}(L) \le \frac{(3/2)^{39/2}}{\sqrt{21870}} = \frac{3^{16}}{2^{20}\sqrt{5}}. \tag{32}$$

The last equality follows from $21870 = 2 \cdot 3^7 \cdot 5$.

This proves an upper bound for the *inherited-radius bare cross-section construction*. The source reports a saturated ambient embedding of its K_9 construction, which would attain this density at the inherited radius. The present audit verifies the abstract matrix and the determinant obstruction; it does not rerun the separate ambient coordinate embedding certificates of that source. Accordingly, the attainment assertion for a specified ambient lattice remains attributed to the supplied manuscript.

If an orthogonal complement happened to have minimum greater than six, one could ask about increasing its packing radius; the fixed-radius formula above does not by itself exclude every gain of that different kind. Nor does it exclude antipode clusters, nonlinear packings, or constructions unrelated to rank-48 sections. The exact conclusion is that one cannot improve the source's dimension-39 bare construction by lowering the rank-nine determinant within the stated even integral setting.

15 Detailed audit findings and present replay status

15.1 What was actually repeated on 10 September 2026

The supplied archive was recovered and its 83 recorded SHA-256 checksums matched before the audit. The released exact-check entry point was rerun and terminated with

```
ALL_RELEASED_EXACT_CERTIFICATES_PASS.
```

The full regeneration entry point was also rerun and terminated with

```
FULL_EXACT_REGENERATION_BYTE_IDENTICAL.
```

This regenerated the rank-two through rank-six representative data and all four rank-seven chunks, together with their merged output. The specified released outputs were byte-identical; separate comparisons also confirmed byte identity of all five raw files in ranks two through six. All 83 manifest entries still matched after the checks. These are completed reruns in this audit, not instructions for a future run.

The runs used CPython 3.12.14 with assertions enabled. Python's `-O` option must not be used, since assertions implement part of the certificate validation. The optional broad E_6 supplementary search was not rerun in full; the necessary E_7 and targeted H_{6318} branch checks were rerun and cover the relevant surviving branches.

The reruns use the released program logic, so they establish reproducible execution and regenerated-list agreement. They are not a wholly independent implementation of the complete rank-seven search. Additional checks were deliberately independent on important smaller components:

- 10036 exact integer-interval checks, 100 independent ellipsoid comparisons, and 60 determinant comparisons;
- direct bounded-coordinate enumeration of the rank-two and rank-three extension stages, recovering 7 and 52 labelled forms without using the released ellipsoid routine;
- independent determinant recomputation of all 138 rank-six parents and 2235 rank-seven children, followed by exact recomputation of the 31 determinant pairs and the $2139 + 96$ split;
- 51 further small ellipsoid comparisons for auxiliary branch arithmetic, verification of the displayed root reflections, and an independent determinant calculation for K_9;
- explicit exhaustive enumeration through norm five for H_{6318}, J_{12393}, J_{13689}, J_{14580}, and $J_{15309,a}, J_{15309,b}$, finding no nonzero vector. This closes an additional validation distinction: a helper that merely finds vectors of norm six establishes their existence, but does not itself exclude shorter vectors;
- the separate 64-tuple proof of the sharpness witness in Section 13, independent of the archived discriminant enumerator.

The finite tests support the audit of the arithmetic interfaces; the general correctness arguments are the explicit formulas and completeness proofs in this document. Random or small-instance tests alone are not offered as a proof that every search is exhaustive.

15.2 Corrections and clarifications

finding	correction and effect on the theorem
Vector versus Gram scaling	The determinant-8192 Gram matrix $4C_{E_7}^{-1}$ represents $2E_7^*$. The earlier label $4E_7^*$ was wrong; the actual matrix, determinant, and exclusion remain correct.
Parent layer in the minimum criterion	The parent must be even integral and have minimum at least six. Nonzero-layer tests do not control its nonzero vectors. These hypotheses held for every actual parent and are now explicit in Proposition 4.1.
Integrality of a constructed extension	Membership $y \in J^*$ guarantees integral cross pairings, but the new norm must also be integral. Even integrality requires $s + \mu_J(a) \in 2\mathbb{Z}$.
Omitted zero-residue layer in a helper	One auxiliary centered-residue test omitted $h^2 s \geq 6$. A real false-positive example exists. The helper was hardened, regression-tested, and rerun; its actual branch output was unchanged. Details follow below.
Norm convention	Minimum always means squared length. Expressions such as $\min L^* = d_{r-1}/d_r$ use that convention throughout.
Dimension-eight equality rigidity	The needed external theorem concerns all Euclidean lattices attaining γ_8, not merely even unimodular lattices. This stronger theorem is stated explicitly.
Root-reflection terminology	The displayed E_7 root Gram matrix is not a simple-root Cartan matrix. Reflections in its displayed roots are checked isometries; their subgroup orbits suffice.
Discrete versus continuous radius	ρ^2_{glue} is a maximum of exact minima over J^*/J. No equality with the ordinary covering radius is needed or asserted.

finding	correction and effect on the theorem
Unrestricted versus flag-compatible children	The two determinant- 13689 children of H_{7722} violate the lifting bound and are not flag-compatible. In contrast, the determinant-14641 children of H_{7986} pass the full shortest-normal test.
Hash validation versus completeness	A matching digest proves file identity. Completeness additionally requires a correct exhaustive generator or independently verified exhaustive proof object; the full generator rerun and its mathematical invariant are reported separately.

These findings should not be hidden behind the phrase "all checks passed." They include a genuine generic programming defect and several mathematically significant missing qualifications. The audit found no failure of the rank-nine lower-bound implication after applying these corrections.

15.3 The zero-residue-layer defect: example, fix, and why the result survives

Let

$$G = \operatorname{diag}(22, 6, 6, 6, 6), \qquad y = (\tfrac{1}{2}, 0, 0, 0, 0)$$

in basis coordinates. The parent is even integral with minimum six. The class $a = y + J$ has order $h = 2$ and minimum $11/2$. Choose $s = 1/2$; the extension has cross column $(11, 0, 0, 0, 0)^{\mathsf{T}}$, last diagonal six, and determinant

$$(22 \cdot 6^4) \cdot \tfrac{1}{2} = 14256.$$

The centered test for $m = 1$ passes, since $s + \mu_J(a) = 1/2 + 11/2 = 6$. But the coordinate vector

$$(-1, 0, 0, 0, 0, 2)$$

in the extension has norm two: its parallel component cancels, and its perpendicular square norm is $h^2 s = 2$. The old generic helper thus accepted a false minimum-six candidate. The corrected version includes

```
if h*h*s < 6: continue.
```

The exact regression constructs the class table for this example and verifies both the old false acceptance and the corrected rejection.

For the actual determinant-7722 branch, every accepted extension has $s = 13689/7722 = 39/22$. Therefore all $|m| \geq 2$ layers have norm at least

$$4s = 78/11 > 6.$$

The omitted inequality is automatically satisfied for nonzero classes of order at least two in the actual surviving output. A class of order one would require $s \geq 6$, which the complete criterion controls. In addition, the relevant target minimum was checked directly, and both unrestricted targets are excluded from the flag by (25). The correction was applied in an isolated hardened copy of the helper; its full branch run passed and produced a transcript byte-identical to the released branch output. The main bottom-up generator uses the complete test $m^2 s < 6$ and did not contain this omission.

15.4 The trust structure of the computer-assisted theorem

It is useful to distinguish four logically different tasks.

1. **Arithmetic validity.** A stored matrix has the stated determinant and minimum; a stored map is unimodular and satisfies its Gram identity; a recorded vector has its asserted norm.
2. **Coverage of finite searches.** Every needed parent, class, even diagonal, layer, and short dual vector was considered. This requires the exhaustive enumeration proof and code, not merely checking the outputs one at a time.
3. **Mathematical reduction.** Every hypothetical counterexample has a flag within the windows, and every retained branch implies the required determinant floor. This is the structural proof in the earlier sections.
4. **Scientific review.** External verification of the complete argument, attribution and novelty, and suitability for publication are additional tasks. An internal replay is not an external referee report or a formal proof in a proof assistant.

The present audit provides new work on the first three tasks, with the reported boundary between full same-code regeneration and independent checks. It does not claim to have completed the fourth.

16 How to reproduce the checks and what each certificate establishes

The released archive has a top-level manifest, readme files in both languages, an assumptions file, a claim-to-certificate map, and a `certificate` directory. Run the checks from an extracted copy so that relative filenames match the documented layout. The primary entry points are

```
sha256sum -c SHA256SUMS
bash run_all_checks.sh
bash run_full_regeneration.sh
```

The mathematical programs use exact integer and rational arithmetic in Python. Keep assertions enabled; do not add `-O` or `-OO` to these certificate runs. The release's runtime requirements and invocation instructions remain authoritative for that archive. For independent work, retaining the original extracted package and writing audit outputs to a separate directory avoids accidentally changing the evidence being checked.

The following map explains the logical job of the main files. Long names are kept exactly so that a reader can locate them.

`exact_lattice.py`
Exact determinants, rational LDL^{T}, short-vector enumeration, signatures, and integral isometry checks.

`extend_flag.py`
Complete discriminant-class enumeration, even diagonal ranges, all necessary layers, parent-level lifting prune, and shortest-normal test.

`verify_lower_flag_chain_exact.py`
Validity of the stored lower raw forms and their representative maps. Coverage comes additionally from regenerating the lower stages with the audited generator.

`verify_regenerated_frozen_bridges_exact.py`
Explicit isometries matching regenerated and frozen representative sets, including all 138 rank-six representatives.

`extend_flag_chunk.py`, `verify_flag_chunks.py`
Generation and checking of the four rank-seven chunks. Parent ranges, input identities, counts, and the merged output must all agree.

`verify_rank7_survivor_linkage_exact.py`
The exact lifting split and all 96 positive links to the six representatives. This checks the bridge from the global search to the individual branch proofs.

`verify_e7_branch_exact.py`
The four rank-eight targets above $\sqrt{3}E_7$, their exhaustive class tables, and their rank-nine determinant floors.

`verify_h6318_branch_exact.py`
All twelve unrestricted children of H_{6318}, their target isometries, and the two target glue radii.

`verify_rank7_topdown.py`
The determinant-8192 obstruction and the determinant-7722 unrestricted branch. Its hardened counterpart in the audit supplement adds the missing zero-residue layer test while preserving the actual branch output.

`verify_rank7_residuals_exact.py`
The determinant-8100 obstruction and the two flag-compatible determinant-14641 children of H_{7986}, followed by their rank-nine floor.

`verify_sqrt3_e8_covering_radius_exact.py`
The discrete discriminant radius 8/3 used to exclude the rank-eight Hermite equality case.

`verify_rank9_extremal_example_exact.py`
The explicit K_9 equality witness, including its determinant, positive definiteness, and minimum.

The supplementary audit contains the corrected helper and its patch, the zero-layer regression and transcript, independent arithmetic checks, theory-window and 64-tuple sharpness checks, and the new replay reports. These supplement the original certificate package; they do not retroactively rename its hashed members. A legacy print label `4 E7^*` in an unchanged old transcript must be read using the explicit scaling correction recorded in this document.

17 What is settled and what remains open

The result established by the audited reduction and exact enumeration is the minimum determinant $e_9(6) = 21870$, with an explicit attaining matrix. It excludes every smaller even integral rank-nine section of a minimum-six ambient lattice and hence every improvement obtained by lowering that determinant in the source's bare dimension-39 construction.

The source's broader questions divide as follows:

question	status of the present work
Least determinant, even integral rank nine, minimum at least six	Determined as 21870 by the computer-assisted theorem.
Uniqueness at determinant 21870	Not proved. The strict-counterexample search does not classify equality cases.
All isometry types in rank nine	Not classified. A bounded flag enumeration used for an obstruction is not a global rank-nine classification.
Primitive embeddings in a specified rank-48 lattice	Not classified or independently rerun here. The abstract obstruction applies to every such embedding if it exists.
Rank ten and the dimension-38 determinant window	Not solved by the rank-nine proof. The source's benchmark and integrality conditions remain a separate starting point.
Antipode or more general clusters in dimensions 38–43	No global optimality result is proved here. The source explicitly leaves room for cluster improvements, including above bare-optimal sections.
Bare statements in dimensions 40–43, ambient constructions, and kissing numbers elsewhere in the source	Quoted for context; the present audit does not constitute a replay or a line-by-line validation of the entire source manuscript.

The final scope is deliberately tied to the universal statement actually proved. A future equality classification must include determinant 21870, retain all equality possibilities in the inequalities, and check their extensions rather than applying the strict bound $d_9 \leq 21868$. A future rank-ten argument requires its own top-level determinant and Hermite analysis. A future cluster argument must also track inter-coset distances and cluster cardinality; a section determinant alone does not determine its density.

A A compact independent verification of the 64-tuple argument

The following code uses only integer arithmetic and directly forms the matrix C appearing in Section 13. It verifies the finite part of (30); the preceding proof explains why checking these tuples is sufficient.

```
from itertools import product
from collections import Counter

C = [
 [ 2,-1,-1,-1,-1,-1,-1, 1],
 [-1, 2, 1, 1, 1, 1, 1, 0],
 [-1, 1, 2, 0, 1, 0, 0,-1],
 [-1, 1, 0, 2, 1, 1, 1, 0],
 [-1, 1, 1, 1, 2, 1, 1,-1],
 [-1, 1, 0, 1, 1, 2, 1, 0],
 [-1, 1, 0, 1, 1, 1, 2, 0],
 [ 1, 0,-1, 0,-1, 0, 0, 2],
]
w = (-1,-1,2,-1,0,0,1,1)
choices = [(-1,2)]*4 + [(0,)]*2 + [(-2,1)]*2

def q(x):
    return sum(x[i]*C[i][j]*x[j]
               for i in range(8) for j in range(8))

counts = Counter(q(r) for r in product(*choices))
assert sum(counts.values()) == 64
assert counts == {
    8:5, 14:9, 20:14, 26:10, 32:8, 38:5,
    44:4, 50:4, 56:1, 62:3, 86:1,
}
assert q(w) == 8 and 2 not in counts
print("K9_RESIDUE_CLASS_MINIMUM_EIGHT")
```

The supplementary verifier additionally checks C^{-1}, positive definiteness, determinant, the exact window endpoints, and the Schur complement using independently written rational arithmetic.

B Glossary of the quantities that must not be confused

notation	meaning
$\min L$	Minimum squared Euclidean length of a nonzero lattice vector.
$\det L$	Determinant of a Gram matrix; the square of the covolume.
$\operatorname{covol} L$	Volume of a fundamental parallelepiped.
L^*	Dual lattice of vectors pairing integrally with L.
$A_L = L^*/L$	Finite discriminant group of an integral lattice, of order $\det L$.
$q_L(a)$	Class norm modulo $2\mathbb{Z}$ for an even lattice.
$\mu_L(a)$	Exact smallest norm in a discriminant class.
$\rho_{\mathrm{glue}}(L)^2$	Maximum of $\mu_L(a)$ over the finite group; the square of the discrete glue radius.
$R(L)$	Ordinary continuous covering radius of all Euclidean space.
s	Squared perpendicular length, equivalently the Schur complement or the ratio of child determinant to parent determinant.
$h = \operatorname{ord}(a)$	Order of an extension class; the nonzero zero-class layer first occurs at index h.
γ_r	Hermite constant for all rank-r Euclidean lattices, without integrality restrictions.
d_r	Determinant of rank r in a chosen saturated shortest-dual flag.
Labelled child	An extension presentation tied to a chosen parent representative and class; different presentations may be isometric.
Flag-compatible child	A child whose perpendicular parent normal is shortest in the child dual; passing a necessary lifting bound alone does not prove this.
$e_9(6)$	Minimum determinant over all positive-definite even integral rank-nine lattices of minimum at least six.

References

[1] I. Dorofeev, X. Sun, and C. Wang, *Optimal Extensions of Cross-Sections: Sphere Packings in Dimensions 38 to 43*, supplied confidential manuscript, 2026. The relevant open questions are quoted in Section 1. No public availability is implied.

[2] *Rank-nine complete certificates*, confidential accompanying exact arithmetic package, recovered release with 83-entry SHA-256 manifest; full verification and regeneration repeated on 10 September 2026. Supplementary audit files contain the corrected generic helper, regression, independent arithmetic checks, and new transcripts.

[3] J. H. Conway and N. J. A. Sloane, *Sphere Packings, Lattices and Groups*, third edition, Springer, 1999. General background on Gram matrices, duality, discriminant groups, root lattices, and Hermite constants.

[4] J. Martinet, *Perfect Lattices in Euclidean Spaces*, Springer, 2003. Background on extreme lattices and low-dimensional Hermite theory.

[5] H. F. Blichfeldt, *The minimum values of positive quadratic forms in six, seven and eight variables*, Mathematische Zeitschrift **39** (1935), 1–15. https://doi.org/10.1007/BF01201341. The publisher metadata was checked during this audit; the audit does not claim to have independently rederived all results in that paper.

[6] J. Wen and X.-W. Chang, *Sharper Bounds on Four Lattice Constants*, arXiv:2204.08021, 2022, Table 1. https://arxiv.org/abs/2204.08021. An accessible explicit table of γ_r for $1 \le r \le 8$.

[7] M. S. Viazovska, *The sphere packing problem in dimension 8*, Annals of Mathematics **185** (2017), 991–1015. https://doi.org/10.4007/annals.2017.185.3.7. Author manuscript: https://arxiv.org/abs/1603.04246. The statement following Theorem 1 gives uniqueness among periodic optimal packings, supplying the all-lattice equality rigidity used here.

[8] R. Bacher, *A new inequality for the Hermite constants*, arXiv:math/0603477, 2006. https://arxiv.org/abs/math/0603477. Mordell's inequality is recalled there; the needed inequality is proved directly in Section 6 of the present note.